SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Project Executable Files

Date	07 July 2026
Team ID	
Project Name	HDI Predictor using Machine Learning and Flask
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes (Located in HDI/)
2	README / Setup Guide (instructions to run the project)	Yes (See README.md)
3	requirements.txt / package.json / dependency file	Yes (Dependencies listed in README.md)
4	Database schema / seed files (if applicable)	Yes (CSV-based database HDI_Cleaned.csv)
5	Environment configuration file (.env.example or similar)	NA (Configuration is handled directly via app parameters)
6	Deployed application URL (if hosted)	No (Local execution setup)
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA (Web application)
8	Dockerfile / Containerization config (if applicable)	NA (Python virtual environment setup)
9	Test files and test results	Yes (Located in 6.Project Testing/)
10	Demo video or walkthrough (if required)	Yes (Documentation in 8.Project

		Demonstration/)

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

```
None
HDI/
├── 1. Brainstorming & Ideation/
├── 2. Requirement Analysis/
├── 3. Project Design Phase/
├── 4. Project Planning Phase/
├── 5. Project Development Phase/
│   └── Human Development Index Code Files/
│       ├── Dataset/
│       │   ├── HDI.csv # Raw dataset
│       │   └── HDI_Cleaned.csv # Cleaned/preprocessed dataset
│       ├── Flask/
│       │   ├── app.py # Web App Entrypoint
│       │   ├── HDI.pkl # Pickled Linear Regression Model
│       │   ├── ml/
│       │   │   └── predictor.py # Core ML inference code
│       │   ├── routes/
│       │   │   ├── home.py
│       │   │   ├── predict.py
│       │   │   └── about.py
│       │   ├── static/ # Stylesheet & assets
│       │   └── templates/ # Jinja2 galaxy-themed views
│       └── Training/
│           └── HumDevIndex.ipynb # Model training notebook
├── 6.Project Testing/
├── 7.Project Documentation/
└── 8.Project Demonstration/
```

Step 3: Deployment / Access Details

Field	**Details**
Hosted / Deployed URL	Run Locally (http://localhost:5000)
Login Credentials (Demo)	No authentication required (Open Access)
Platform / Hosting Provider	Local Host (Python/Flask)
Repository Link	https://github.com/pspvv/HDI.git

Demo Video Link	https://youtu.be/ZoBjbXo-wmY

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:
Pre-requisites:

- Python 3.x
- Flask, scikit-learn, pandas, numpy, matplotlib, seaborn

Steps:

1. Step 1: Navigate to the workspace: `cd HDI`
2. Step 2: Set up a Virtual Environment: `python -m venv venv` (On Windows: `venv\Scripts\activate`; On macOS/Linux: `source venv/bin/activate`)
3. Step 3: Install Dependencies: `pip install flask scikit-learn pandas numpy matplotlib seaborn`
4. Step 4: Run the Application: `cd "5. Project Development Phase\Human Development Index Code Files\Flask"` then `python app.py`
5. Step 5: Access Application: Open `http://localhost:5000` in your web browser.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	File uploads must exactly match expected CSV headers for Batch Predictions	Sample CSV formats are provided in the UI under Batch Prediction instructions
2	Out-of-bounds indicators may output distorted prediction scores	The application checks standard input ranges and alerts the user of potential outliers